Finite-energy extension from every level-l half Sierpiński gasket

Partial-result packet for arXiv:1702.02419

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Status: candidate partial result, likely valid. For every $l \geq 2$, reflection gives a bounded linear extension of every finite-energy function on the vertical half domain to the whole $\mathcal{SG}_l$, with an exact energy identity. The source's upper/lower-domain and higher-Sobolev extension problems remain open.

1 Source problem

Cao and Qiu study half, upper, and lower domains in the level-l Sierpiński gaskets and ask on PDF page 6 how to extend a finite-energy function from such a domain to the whole gasket [1, p. 6]. Their exact sentence is:

> "One could also consider how to extend a function of finite energy defined on Ω to a function of finite energy on the whole $\mathcal{SG}_l$, and analogous problems for other Sobolev spaces."

The source passage is reproduced below.

> At the end of this section, we list some previous work on related topics. See [GKQS], [HKu], [J], [Ki4], [LRSU], [LS], [OS] and the references therein. In particular, in [LRSU], some extension problems on $\mathcal{SG}$ are studied which is to find a function with certain prescribed data such as values and derivatives at a finite set, that minimizes prescribed Sobolev types of norms. It is interesting to consider analogous problems on domains in this paper. We leave these as open problems for future research. The energy estimates considered in this paper characterize the restriction to the Cantor set boundary X of functions of finite energy on Ω. It is also interesting to characterize the traces on X of functions in some other Sobolev spaces, such as $dom_{L^2}(\Delta^k)$ defined as $\{u \in L^2(\mathcal{SG}_l) : \Delta^j u \in L^2(\mathcal{SG}_l), \forall j \leq k\}$. One could also consider how to extend a function of finite energy defined on Ω to a function of finite energy on the whole $\mathcal{SG}_l$, and analogous problems for other Sobolev spaces. Related problems are discussed in [LS], [GKQS]. The above men-

The theorem below resolves the finite-energy question for the full vertical half-domain family. It is stronger than a harmonic-extension statement: the input function is arbitrary in the local resistance-form domain.

2 Definitions and statement

Let ρ be reflection in the vertical symmetry axis of $\mathcal{SG}_l$. Let h_a be the antisymmetric harmonic function used by Cao–Qiu, normalized by $h_a(q_0) = 0$, $h_a(q_1) = 1$, and $h_a(q_2) = -1$. Put

$$\Omega^- = \{h_a > 0\}, \qquad X = \{h_a = 0\}, \qquad \Omega^+ = \rho(\Omega^-) = \{h_a < 0\}.$$

Thus X is the pointwise fixed set of ρ, and $\mathcal{SG}_l = \Omega^- \sqcup X \sqcup \Omega^+$.

Write $\Gamma_m = (V_m, E_m)$ for the usual level-m graph approximation and r_l for the resistance renormalization factor. The local graph energy on the closed left half is

$$\mathcal{E}_{m,-}(u) = r_l^{-m} \sum_{e=\{x,y\}\in E_m^-} (u(x) - u(y))^2,$$

where E_m^- is the left member of every non-fixed reflection orbit of graph edges. Equivalently, these are the edge terms in the standard left-half exhaustion used for $\mathcal{E}_{\Omega^-}$ in the source. Let

$$\mathcal{F}(\overline{\Omega^-}) = \{u \in C(\overline{\Omega^-}) : \mathcal{E}_{\Omega^-}(u) := \lim_m \mathcal{E}_{m,-}(u) < \infty\}.$$

Proof intuition

At each graph level, reflection partitions the non-fixed edges into pairs. An even extension gives the same energy to both edges in a pair. The only edges fixed as sets are horizontal edges crossing the symmetry axis; reflection exchanges their endpoints, so the even extension has equal endpoint values and their energy is zero. The full graph energy is therefore exactly twice the left-half energy at every finite level. Passing to the resistance-form limit simultaneously proves existence, boundedness, and the sharp identity for this specific extension.

Theorem 1 (even reflection extension)**.** *For every integer $l \geq 2$, define*

$$(\mathsf{R}u)(x) = \begin{cases} u(x), & x \in \overline{\Omega^-}, \\ u(\rho x), & x \in \Omega^+. \end{cases}$$

Then

$$\mathsf{R} : \mathcal{F}(\overline{\Omega^-}) \longrightarrow \mathcal{F}(\mathcal{SG}_l)$$

is a linear right inverse to restriction, and

$$\mathcal{E}_{\mathcal{SG}_l}(\mathsf{R}u) = 2\mathcal{E}_{\Omega^-}(u). \tag{1}$$

It also preserves the uniform norm: $\|\mathsf{R}u\|_{\infty,\mathcal{SG}_l} = \|u\|_{\infty,\overline{\Omega^-}}$. Consequently, with the anchored resistance norm

$$\|u\|_{*,Y}^2 = |u(q_0)|^2 + \mathcal{E}_Y(u),$$

the extension operator has norm at most $\sqrt{2}$.

3 Proof

The reflection ρ permutes the contraction maps defining $\mathcal{SG}_l$ and is therefore an automorphism of every graph Γ_m. Hence it permutes E_m. The triangular lattice has no nontrivial edge contained in the vertical axis. Every edge orbit consequently has one of two forms:

1. $\{e, \rho e\}$, where e is on the left and ρe is on the right;
2. a singleton $\{e\}$, where $e = \{x, \rho x\}$ crosses the axis and reflection exchanges its endpoints.

This classification also follows cell by cell: the three graph-edge directions are horizontal and at angles $\pm\pi/3$, whereas the fixed axis is vertical.

Because ρ fixes X pointwise, the two formulas defining $\mathsf{R}u$ agree on X. They therefore define a continuous function on $\mathcal{SG}_l$. For a paired orbit, say $e = \{x, y\}$ on the left, the two contributions to the graph energy of $\mathsf{R}u$ are equal:

$$\big((\mathsf{R}u)(x) - (\mathsf{R}u)(y)\big)^2 = \big((\mathsf{R}u)(\rho x) - (\mathsf{R}u)(\rho y)\big)^2 = (u(x) - u(y))^2.$$

For a singleton orbit $e = \{x, \rho x\}$, evenness gives $(\mathsf{R}u)(x) = (\mathsf{R}u)(\rho x)$, so its contribution is zero. Summing edge orbits and inserting the common factor r_l^{-m} gives the exact finite-level identity

$$\mathcal{E}_m(\mathsf{R}u) = 2\mathcal{E}_{m,-}(u) \qquad (m \geq 0).$$

Taking increasing limits proves (1); in particular $\mathsf{R}u$ has finite global energy. Linearity, the restriction identity, and the uniform norm identity are immediate from the definition. Finally,

$$\|\mathsf{R}u\|_{*,\mathcal{SG}_l}^2 = |u(q_0)|^2 + 2\mathcal{E}_{\Omega^-}(u) \leq 2\big(|u(q_0)|^2 + \mathcal{E}_{\Omega^-}(u)\big),$$

which proves the stated norm bound. □

Corollary 2 (energy-minimizing completion of the other half)**.** *For each $u \in \mathcal{F}(\overline{\Omega^-})$, there is a global finite-energy extension that agrees with u on $\overline{\Omega^-}$ and is harmonic on Ω^+. Its total energy is at most $2\mathcal{E}_{\Omega^-}(u)$.*

Proof. The reflected function $\mathsf{R}u$ supplies a finite-energy competitor on Ω^+ with boundary trace $u|_X$. Minimizing the right-half energy with that trace, by the usual Dirichlet principle for the resistance form (the same principle used in Proposition 1.1 of [1]), gives a harmonic completion whose right-half energy is no larger than that of the reflected competitor, namely $\mathcal{E}_{\Omega^-}(u)$. Add the unchanged left-half energy. □

4 Verification, scope, and novelty audit

The edge-orbit identity was checked independently by `code/verify_reflection_energy.py`. The script constructs the standard triangular IFS graphs for $2 \leq l \leq 6$, through depth three for $l \leq 4$ and depth two otherwise, verifies that reflection is an edge automorphism and that no nontrivial edge lies in the fixed axis, and tests the energy identity on seeded random symmetric vertex data. This is a finite regression check only; the edge-orbit argument above is the proof.

The result is deliberately scoped to the vertical half domains and the first-order resistance energy. It does not settle arbitrary horizontal upper or lower cuts, nor extension in $\mathrm{dom}_{L^2}(\Delta^k)$. Even reflection also need not make a harmonic function globally harmonic across X: a normal-flux matching condition can fail.

The run searched its registry, solution, attempt, and proof-gap indexes for arXiv:1702.02419 and the core extension terms. Bounded primary-source searches for half-gasket finite-energy reflection/extension recovered Li–Strichartz's $\mathcal{SG}_2$ work on trace spaces and piecewise-biharmonic extensions for Laplacian domains [2], Cao–Qiu's later Sobolev work on p.c.f. sets [3], and a bottom-line trace theorem [4], but no all-l statement of Theorem 1. Because the symmetry argument is elementary, the novelty claim is plausible but especially provisional: the theorem may be folklore or implicit in resistance-form locality.

Human-review focus. Confirm that the paper's local energy $\mathcal{E}_{\Omega^-}$ agrees with the left-edge exhaustion used here for all l, especially at graph edges setwise fixed by reflection; then check the bounded literature search for an implicit reflection extension lemma.

References

[1] S. Cao and H. Qiu, *Boundary Value Problems for harmonic functions on domains in Sierpiński gaskets*, arXiv:1702.02419 (2017).

[2] W. Li and R. S. Strichartz, *Boundary Value Problems on a Half Sierpiński Gasket*, J. Fractal Geom. 1 (2014), 1–43; arXiv:1304.3522.

[3] S. Cao and H. Qiu, *Sobolev spaces on p.c.f. self-similar sets: critical orders and atomic decompositions*, arXiv:1904.00342 (2020).

[4] S. Cao, S. Li, R. S. Strichartz, and P. Talwai, *A Trace Theorem For Sobolev Spaces On The Sierpiński Gasket*, arXiv:1905.03391 (2019).